

Refilling the Buffer in a Storm

The Netherlands' countercyclical capital buffer, 0% → 2% (2022–2024)

Zheng Huang · Diego Gutiérrez

Micro- & Macroprudential Management · Country implementation of a CCyB increase

Master of Financial Technology · Frankfurt School of Finance & Management

October 2026

SAY

- Good morning. We are Zheng Huang and Diego Gutiérrez, and our case is the Netherlands.
- In 2022 and 2023 the Dutch central bank, De Nederlandsche Bank or DNB, raised its countercyclical capital buffer from zero to two percent. We want to explain why it did that, and whether it was a good idea.

NUMBERS

- CCyB 0% → 2% (fully binding 31 May 2024)

TERMS EXPLAINED

- **DNB:** De Nederlandsche Bank, the Dutch central bank and macroprudential authority.

In 2022–23 the Netherlands raised its countercyclical capital buffer (CCyB) to 2% while the rulebook said 0%. It was refilling a buffer, not fighting a boom

SUMMARY

De Nederlandsche Bank (DNB) raised the CCyB from 0% to 1% (announced May 2022) and then to 2% (announced May 2023, binding since 31 May 2024). At both decisions the Basel reference indicator, the credit-to-GDP gap, stood far below zero (−33 and −47 percentage points), so the rulebook pointed to a buffer of 0%.

Q1 · ENVIRONMENT

The economy had recovered from COVID, but war, an energy shock and 11.6% inflation darkened the outlook, and the ECB began raising rates in July 2022. Credit was cold, housing was hot and then cooled, and the banks were strong and growing more profitable.

Q2 · CRITERIA AND RISKS

DNB's 2022 framework sets 2% as the normal level, calibrated on past crisis losses rather than on credit growth. The buffer insures against shocks that no indicator predicts, while housing risks are handled by other tools. It also kept a promise made in March 2020.

Q3 · IMPLICATIONS

The buffer was cheap, at most about a fifth of the big banks' spare capital. It was largely a swap for cuts in structural buffers, and we see no sign of a contraction in lending. The Netherlands now has €6.7bn it can release.

Our verdict: **normalisation, not tightening**. The price is a greater reliance on DNB's judgement, and the buffer has not yet been tested in a crisis.

Source: DNB (2020, 2022, 2023, 2026); DNB Financial Stability Reports 2022–23; BIS credit-to-GDP gaps; ECB; bank disclosures; own calculations.
Huang & Gutiérrez · The Netherlands' CCyB

2/26

SAY

- Here is the whole story on one slide. DNB raised the buffer in two steps, to one percent and then to two percent. Yet at both decisions the indicator at the centre of the Basel rules, the credit-to-GDP gap, was deeply negative, so the rulebook said the buffer should be zero.
- The three columns follow the three questions of the assignment: the environment, the criteria and risks, and the implications.
- Our verdict is that this was **normalisation, not tightening**. DNB was refilling an empty buffer, not fighting a credit boom. The price is that the policy relies on DNB's judgement rather than on a rule.

NUMBERS

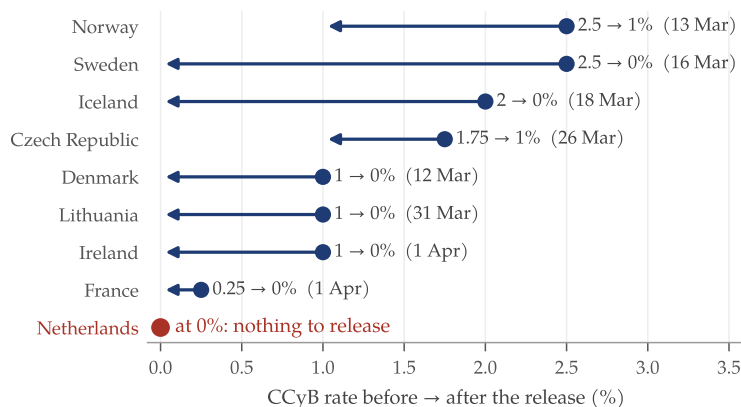
- Announcements: May 2022 (1%) and May 2023 (2%); binding 25 May 2023 and 31 May 2024
- Credit-to-GDP gap at the decisions: −33pp and −47pp → Basel guide 0%
- Inflation 11.6% (2022) · CET1 17.7% (end-2021) · €6.7bn releasable

TERMS EXPLAINED

- **CCyB:** Countercyclical capital buffer. Extra bank capital (0–2.5% of risk-weighted assets) built in good times, released in bad times.
- **Credit-to-GDP gap:** How far credit/GDP is above its long-run trend; the Basel starting point for the CCyB.
- **O-SII:** "Other systemically important institution". The O-SII buffer is an extra capital charge on large banks.

In March 2020 COVID struck and neighbours released their buffers, but the Dutch buffer was empty

PROLOGUE



Within three weeks, eight European countries cut their countercyclical buffers so that banks could absorb losses and keep lending. Belgium and Germany cancelled increases they had already announced.

The Netherlands had nothing to release. DNB improvised: on 17 March 2020 it cut the systemic buffers of ING, Rabobank and ABN AMRO and postponed a planned floor on mortgage risk weights. This freed €8bn of capital, enough to support up to €200bn of lending.

"Once the situation is back to normal, DNB will compensate the systemic buffers reduction by gradually increasing the countercyclical capital buffer to 2%."

DNB, 17 March 2020

Source: ESRB, CCyB rates table (announcements March–April 2020); DNB press release "DNB lowers bank buffer requirements to support lending", 17 Mar 2020.
Huang & Gutiérrez · The Netherlands' CCyB

3/26

SAY

- The story starts in March 2020. When COVID hit, our neighbours reacted within three weeks: eight European countries cut their countercyclical buffers, as the chart shows. Sweden and Norway had been at two and a half percent; Denmark, Ireland and Lithuania at one percent. Buffers built in good times were used exactly as intended.
- The Netherlands was at **zero**, the red dot at the bottom. There was nothing to release.
- So DNB improvised. It cut the systemic buffers of ING, Rabobank and ABN AMRO and postponed a planned floor on mortgage risk weights. This freed about eight billion euros of capital, enough to support up to 200 billion euros of lending. But those buffers were never designed to be released.
- And DNB made a promise (*pause*): once things were back to normal, it would rebuild this capital as a two-percent countercyclical buffer.

NUMBERS

- Releases (rate in force → new rate):
 - Denmark 1.0% → 0% (12 Mar)
 - Norway 2.5% → 1.0% (13 Mar)
 - Sweden 2.5% → 0% (16 Mar)
 - Lithuania 1.0% → 0% (31 Mar)
 - France 0.25% → 0% (1 Apr)
 - Ireland 1.0% → 0% (1 Apr)
- Cancelled pending increases: Belgium 0.5% (27 Mar) · Germany 0.25% (31 Mar)
- Also: Iceland 2.0% → 0% · Czech Republic 1.75% → 1.0%
- NL: systemic buffers 3% → 2.5% (ING) / 2% (Rabo) / 1.5% (ABN AMRO); €8bn freed (incl. the floor postponement); up to €200bn of lending

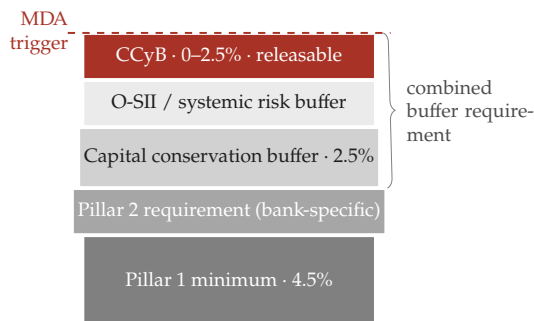
TERMS EXPLAINED

- **Release:** The authority lowers the CCyB rate with immediate effect, so banks may use that capital to absorb losses and keep lending.
- **Systemic (risk) buffer:** A structural capital charge on large banks; not designed to be released.
- **Mortgage risk-weight floor:** A minimum average risk weight on Dutch mortgages for banks using internal models.

NEXT → "That empty buffer mattered, because not all capital can be released."

That mattered because the CCyB is the only layer of bank capital an authority can release in a crisis

PROLOGUE



Capital stack in % of risk-weighted assets (RWA), not to scale.

Read the stack from the bottom. Banks must always meet the minimum requirements; the buffers on top can be used in bad times.

If a bank's capital falls below the top of its buffers (the MDA trigger), its dividends and bonuses are restricted automatically. Banks therefore avoid using buffers even when they are allowed to. In COVID, banks close to this line cut lending instead.

The CCyB works differently. The authority can switch it off, so the requirement itself falls and the bank faces no penalty and no stigma. Increases bind only after twelve months, but a release is immediate.

Only a buffer that the authority can **release** is truly usable in a crisis, and it has to be filled beforehand.

Source: CRD (Arts. 130–136); BCBS (2024) d585; Couaillier, Lo Duca, Reghezza & Rodriguez d'Acari (2022), ECB WP 2644.
Huang & Gutiérrez · The Netherlands' CCyB

4/26

SAY

- Why did an empty buffer matter? Because not all bank capital can be used in a crisis.
- Read the stack from the bottom: first the minimum requirements, then the buffers on top. The red dashed line is the MDA trigger. If a bank falls below it, its dividends and bonuses are restricted automatically, and markets read that as weakness. So in COVID, banks close to this line cut lending rather than use buffers they were formally allowed to use.
- The countercyclical buffer is different. The authority can switch it off, so the requirement itself falls, with no penalty and no stigma. It is the only truly **releasable** layer. And while an increase binds only after twelve months, a release is immediate, so the buffer must be filled before a crisis.

NUMBERS

- Pillar 1 minimum 4.5% CET1 · capital conservation buffer 2.5%
- CCyB 0–2.5% of RWA (higher possible) · increases bind after 12 months · releases are immediate

TERMS EXPLAINED

- **RWA:** Risk-weighted assets. Assets weighted by riskiness (a mortgage counts less than a corporate loan).
- **CET1:** Common Equity Tier 1. The highest-quality capital (shares, retained earnings), as % of RWA.
- **MDA:** Maximum distributable amount. The cap on payouts that applies once capital falls below the combined buffer requirement.
- **Pillar 2 requirement / guidance:** The supervisor's bank-specific add-on (binding) / extra expectation (non-binding).

NEXT → "Two years later DNB refilled exactly that layer, at a strange moment."

Two years later DNB refilled it in the middle of a storm and against its own rulebook. Was that the wrong moment?

THE QUESTION

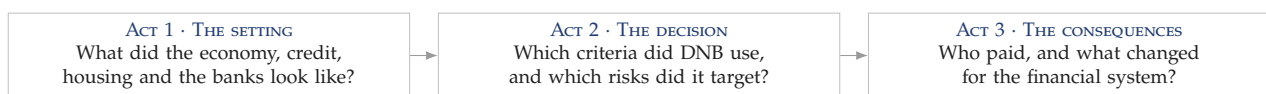
THE STORM OF 2022–23

Russia invaded Ukraine and energy prices hit records. Dutch inflation averaged 11.6% in 2022, the ECB raised rates from July 2022, and house prices were about to fall. The Basel rule, based on the credit gap, said the buffer should be 0%.

THE DECISION

DNB nevertheless raised the CCyB from 0% to 1% (announced May 2022) and then to 2% (announced May 2023). Each step became binding a year later, in May 2023 and May 2024.

Raising capital requirements in a downturn can amplify the cycle, which is exactly the **procyclical mistake** the CCyB was invented to avoid. So was DNB tightening into a slowdown, fighting a hidden boom, or doing something else?



Source: CBS (HICP 2022); ECB monetary policy decision, 21 Jul 2022; BIS credit-to-GDP gaps (latest data at each decision: 2021Q4, 2022Q4).
Huang & Gutiérrez · The Netherlands' CCyB

5/26

SAY

- Two years later, in 2022, DNB began to refill the buffer, and the timing looks strange. On the left is the storm: war, record energy prices, 11.6 percent inflation, the first ECB rate hikes and a housing market about to turn. And the Basel indicator said zero. On the right is the decision: one percent, then two percent.
- Raising capital requirements in a downturn can amplify the cycle. That is exactly the procyclical mistake the buffer was invented to avoid.
- So our question is: **was this the wrong moment?** Was DNB tightening into a slowdown, fighting a hidden boom, or doing something else? We answer in three acts, one for each question of the assignment.

NUMBERS

- HICP inflation 11.6% (2022) · ECB first hike 21 Jul 2022 (+50bp)
- Credit-to-GDP gap –33pp (data to 2021Q4) and –47pp (data to 2022Q4)

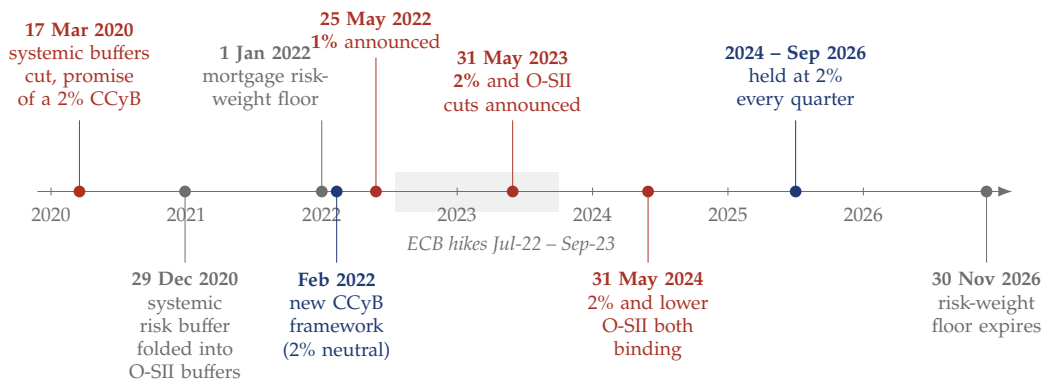
TERMS EXPLAINED

- **Procyclical:** Policy that reinforces the cycle (tighter in bad times, looser in good times).
- **Basel buffer guide:** Gap below 2pp → 0% buffer; above 10pp → 2.5%; linear in between.

NEXT → "Act 1, the setting. The refill followed a path promised in 2020."

The refill followed the path promised in 2020, one step per year, just as the ECB started to tighten

Q1 · ENVIRONMENT



DNB moved in two equal steps of one percentage point, a year apart, and then stopped at 2%. The first step was announced two months before the ECB's first rate hike.

Source: DNB press releases (17 Mar 2020; 27 May 2022; 31 May 2023; 24 Apr 2026; 17 Sep 2026); ESRB O-SII notifications (27 Nov 2020; 31 May 2023); ECB (21 Jul 2022; 14 Sep 2023).

Huang & Gutiérrez · The Netherlands' CCyB

6/26

SAY

- Act one is the setting, and it starts with the timeline. In March 2020 DNB cut the systemic buffers and made its promise. In February 2022 it published a framework with two percent as the normal level. In May 2022 it announced one percent, two months before the ECB's first hike. In May 2023 it announced two percent, together with lower buffers for the largest banks, and both became binding in May 2024.
- Since then DNB has held two percent at every quarterly review. So the path was two equal steps, a year apart, and then a stop. Keep that pattern in mind for our verdict.

NUMBERS

- 17 Mar 2020 · 29 Dec 2020 · 1 Jan 2022 · Feb 2022 · 25 May 2022 · 31 May 2023 · 31 May 2024 · 30 Nov 2026
- ECB hikes: first on 21 Jul 2022; still hiking on 14 Sep 2023

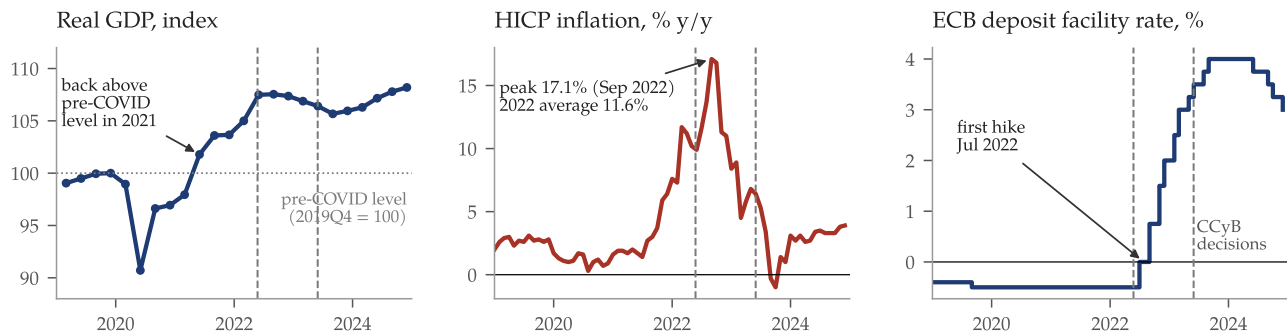
TERMS EXPLAINED

- **Announcement vs binding date:** Banks get 12 months between a CCyB increase being announced and having to meet it.
- **CRD V:** The 2019 revision of the EU Capital Requirements Directive; it made the O-SII and systemic risk buffers additive, prompting DNB's conversion.

NEXT → "By then the economy had fully recovered, but the outlook was darkening."

By then the economy had fully recovered, but war, energy prices and inflation were darkening the outlook

Q1 · ENVIRONMENT



GDP was back above its pre-COVID level in 2021, which is the condition DNB's framework sets before a buffer is built. Then inflation rose to a peak of 17.1%, and the ECB moved its deposit rate from -0.5% to 4% in fourteen months. DNB wrote in spring 2022 that *"the economic outlook has worsened due to the war in Ukraine and high inflation."*

The macro picture argued both ways: build the buffer while the economy can take it, or wait because it is weakening.

Source: ECB Data Portal (Dutch real GDP, MNA; Dutch HICP, ICP; deposit facility rate, FM), current vintage; DNB Financial Stability Report, spring 2022. Dashed lines: CCyB announcements.

Huang & Gutiérrez · The Netherlands' CCyB

7/26

SAY

- What did the economy look like? The left panel shows that real GDP was back above its pre-COVID level during 2021, which is the condition DNB's framework sets before rebuilding the buffer.
- Then the storm arrived. Inflation, in the middle, peaked at 17 percent in September 2022, and on the right the ECB moved its deposit rate from minus half a percent to four percent in fourteen months. DNB warned that higher funding costs could strain the debt of governments, firms and households.
- So the macro picture argued **both ways**: build while the economy can take it, or wait because it is weakening.

NUMBERS

- GDP above pre-COVID level in 2021 (CBS first estimate: 2021Q3; current ECB vintage: 2021Q2, $+3.7\%$ vs 2019Q4 by end-2021)
- HICP 11.6% average in 2022, peak 17.1% (Sep 2022); national CPI 10.0%
- ECB deposit rate $-0.50\% \rightarrow 4.00\%$ (Jul 2022 – Sep 2023); first hike $+50\text{bp}$ on 21 Jul 2022

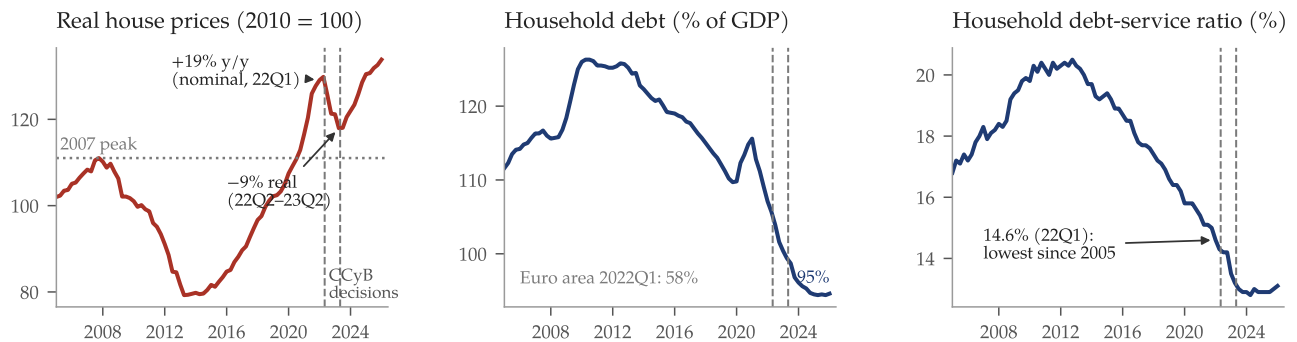
TERMS EXPLAINED

- **HICP:** Harmonised Index of Consumer Prices, the euro-area-comparable inflation measure.
- **Basis point (bp):** 0.01 percentage point; $50\text{bp} = 0.5\text{pp}$.

NEXT → "In the financial system, credit was cold and only housing was hot."

In the financial system credit was cold and only housing was hot, and even housing soon turned

Q1 · ENVIRONMENT



Housing was hot. Nominal prices rose 19% in the year to early 2022, and DNB itself wrote that *"the overheated housing market calls for further measures."* As mortgage rates rose, real prices fell by 9%.

Credit was not. Household debt is high (107% of GDP against 58% in the euro area) but it was falling, and the share of income spent on debt service was the lowest since 2005.

If there was a boom, it was in house prices, not in credit.

Source: BIS (WS_SPP, WS_TC, WS_DSR); DNB Financial Stability Report, spring 2022; own calculations. Dashed lines: CCyB announcements (May 2022, May 2023). Huang & Gutiérrez · The Netherlands' CCyB

8/26

SAY

- What about the financial system? On the left you see housing, and housing was hot. Nominal house prices rose 19 percent in a single year, real prices were well above their 2007 peak, and DNB itself called the market overheated. But as mortgage rates rose, real prices fell by about nine percent.
- In the middle is household debt. It is high, at 107 percent of GDP against 58 percent in the euro area, but it was **falling**. On the right is the share of income that households spend on interest and repayments. It was the lowest since 2005.
- So if there was a boom, it was in house prices, not in credit.

NUMBERS

- Nominal house prices +19.0% y/y (2022Q1); real index 129 vs 2007 peak 111 (+16%)
- Real house prices -9.2% (2022Q2-2023Q2)
- Household debt/GDP: 110% (2019Q4) → 107% (2022Q1) → 100% (2023Q1); euro area 57.7% (2022Q1)
- Debt-service ratio, households: 14.6% (2022Q1)

TERMS EXPLAINED

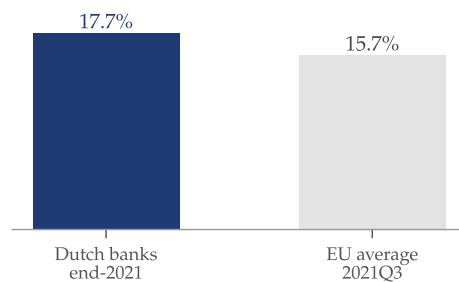
- **Debt-service ratio:** Interest plus principal payments as a share of income.
- **Real house prices:** Nominal prices deflated by consumer prices.

NEXT → "And the banks entered this storm well capitalised."

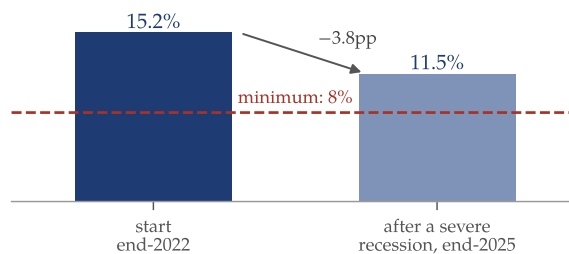
Banks entered this storm well capitalised, and rising rates were lifting their profits

Q1 · ENVIRONMENT

Core capital (CET1) ratio, % of RWA



DNB stress test, four largest banks' CET1 ratio



Dutch banks held more core capital than the EU average, and in DNB's stress test the four largest banks stayed well above the minimum after a severe recession. Rising rates also lifted their net interest income, by 6.7% in 2022; DNB had expected higher rates to *"ultimately have a positive impact on the profitability of financial institutions."*

Act 1 in one sentence: war, inflation, rate hikes and a turning housing market argued against raising the buffer, while a recovered economy, cold credit and strong banks argued for it.

Source: DNB Financial Stability Reports, spring 2022 (CET1 17.7%, EU 15.7%) and spring 2023 (stress test, net interest income).
Huang & Gutiérrez · The Netherlands' CCyB

9/26

SAY

- Finally, the banks. Dutch banks entered the storm with a core capital ratio of 17.7 percent, two points above the EU average. In DNB's 2023 stress test, the four largest banks fell by 3.8 points in a severe recession but stayed at 11.5 percent, well above the minimum.
- Rising rates also **lifted their profits**: net interest income grew 6.7 percent in 2022. Please remember this, because it will come back in act three.
- So act one in one sentence: war, inflation, rate hikes and a turning housing market argued against raising the buffer, while a recovered economy, cold credit and strong banks argued for it.

NUMBERS

- CET1 17.7% (end-2021; EU 15.7% in 2021Q3) → 16.3% (end-2022, in line with EU)
- Stress test (FSR 2023): 15.2% (end-2022) – 3.8pp → 11.5% at end-2025 (min. 8%)
- Net interest income +6.7% in 2022; 67.7% of bank income

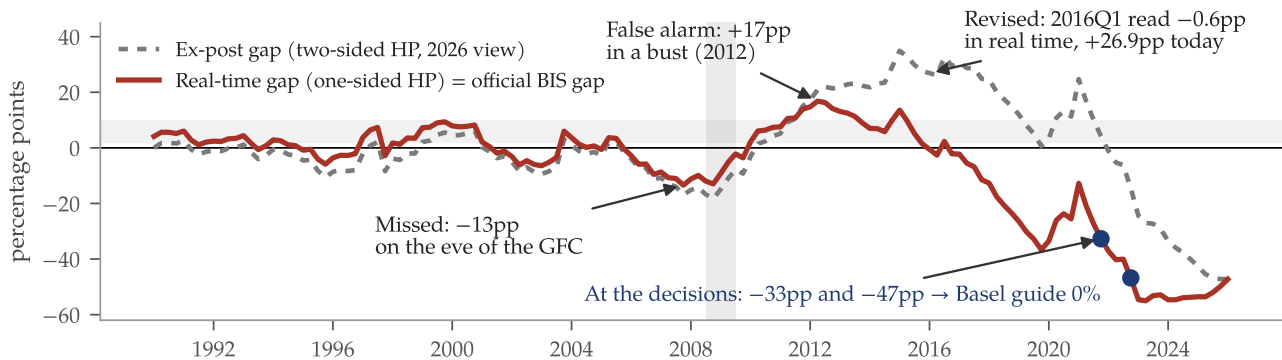
TERMS EXPLAINED

- **Stress test**: A simulation of a severe recession to check whether capital stays above the minimum.
- **Net interest income**: Interest earned on loans minus interest paid on deposits and funding.

NEXT → **HAND-OVER**: "That was the setting. So why did DNB ignore its own rulebook? Diego."

So why ignore the rulebook? On Dutch data the Basel rule had missed the last crisis and raised false alarms

Q2 · CRITERIA & RISKS



The credit-to-GDP gap measures how far credit relative to GDP is above its long-run trend. Under the Basel guide the buffer starts when the gap exceeds 2 points and reaches 2.5% at 10 points.

On Dutch data the gap missed the 2008 crisis, sounded a false alarm in 2012 and was later revised. At the two decisions it read -33 and -47 points.

Source: BIS credit-to-GDP gaps (Q.NL.P.A); Drehmann et al. (2010); Edge & Meisenzahl (2011); own one-sided HP filter ($\lambda = 400,000$).
Huang & Gutiérrez · The Netherlands' CCyB

10/26

SAY

- Thank you, Zheng. Act two is the decision, and the first question is: why did DNB ignore the rulebook? Because on Dutch data the rulebook has a poor record.
- The red line is the credit-to-GDP gap: how far credit relative to GDP is above its long-run trend. Under the Basel guide the buffer starts when the gap exceeds two points. We rebuilt the gap ourselves, and it matches the official BIS series almost exactly.
- Look at its record. It **missed** the financial crisis, reading minus 13 on the eve of 2008. It raised a **false alarm** in 2012, in the middle of a bust. And it gets **revised**: 2016 read slightly negative at the time and plus 27 with today's data. The reason is that Dutch credit grew from about 50 to 350 percent of GDP, and the filter mistakes that structural rise for a trend.
- At the two decisions the gap read minus 33 and minus 47. Taken literally, the rule said zero.

NUMBERS

- Gap: -13.4pp (2007Q4) · +16.8pp (2012Q2) · 2016Q1 -0.6pp real time vs +26.9pp ex post
- -32.7pp (2021Q4) and -46.8pp (2022Q4) at the decisions
- Credit/GDP 47% (1961) → 352% (2015Q1 peak)
- Replication error ≤ 0.00005 pp (1971–2026)

TERMS EXPLAINED

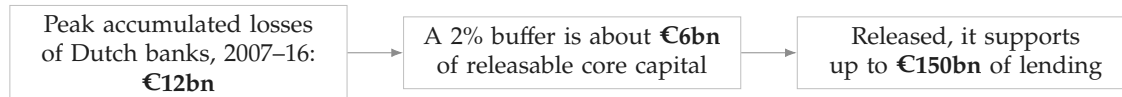
- **HP filter:** Splits a series into a smooth trend and a cycle; $\lambda = 400,000$ for credit cycles.
- **One-sided vs two-sided filter:** One-sided uses only data up to each date (what policymakers see); two-sided also uses later data (hindsight).
- **False alarm / missed crisis:** Type II / type I errors of an early-warning indicator.

DNB used its own compass instead: 2% in normal times, calibrated on past crisis losses rather than on credit growth

Q2 · CRITERIA & RISKS



built up by 1pp a year



The framework sets the buffer at 2% when risks are neither high nor low, a so-called *positive neutral* buffer. DNB reads a dashboard of indicators but stresses that there is “no mechanical link . . . guided discretion.” Seventeen jurisdictions now work this way, and Sweden, the UK and Poland also target 2%.

The 2% is sized to absorb about half of past crisis losses. It does not depend on credit growth.

Source: DNB (2022), Analytical framework for setting the CCyB in the Netherlands; BCBS (2024) d585, Table 1.
Huang & Gutiérrez · The Netherlands' CCyB

11/26

SAY

- So DNB used its own compass. Its 2022 framework has four phases: release after a crisis, **two percent in normal times**, more than two percent when risks are clearly elevated, and release again in a crisis. This is a *positive neutral* buffer: it is already filled when risks are neither high nor low.
- The two percent comes from history, not from credit growth. At the peak, past crises cost Dutch banks about 12 billion euros. Two percent is about six billion euros of releasable capital, roughly half of those losses, and enough to support up to 150 billion euros of lending. The ECB's own estimates range from 1.1 to 1.8 percent, so two percent is at the top of the range, but not outside it.
- Seventeen jurisdictions now work this way, and Sweden, the UK and Poland also target two percent.

NUMBERS

- Peak accumulated losses €12bn (2007–2016) → 2% ≈ €6bn → up to €150bn of lending
- Build-up: 1pp per year → 2% after two years
- 17 jurisdictions with a positive neutral CCyB (BCBS 2024); ECB loss-based range 1.1–1.8%

TERMS EXPLAINED

- **Positive neutral CCyB:** A CCyB kept above zero in a “standard” risk environment.
- **Peak accumulated losses:** The largest cumulative loss a bank suffered over a crisis period.
- **Guided discretion:** Decisions informed by a published set of indicators, with judgement, not a formula.

NEXT → “And its stated reasons were never credit growth.”

Its stated reasons were recovery, uncertainty and the 2020 promise, never credit growth

Q2 · CRITERIA & RISKS

MAY 2022 · FROM 0% TO 1%

*"When the systemic risk buffers . . . were lowered in March 2020, we also announced our intention to **restore the buffers** by raising the CCyB."*

"The risk profile is currently dominated by uncertainty caused by the war in Ukraine, but . . . the robust economic recovery . . . provides grounds for a gradual build-up."

DNB judged cyclical risks to be normal to elevated.

MAY 2023 · FROM 1% TO 2%

*"Although the credit-to-GDP gap shows **no signs of excessive credit growth**, some other indicators already point to higher cyclical risk."*

DNB cited investors' rising appetite for risk, falling real-estate prices, the debt sustainability of firms and governments, and banks that were "robust, partly due to rising interest rates."

Both decisions rested on the same three pillars: a normal-to-elevated risk picture, strong banks and the 2020 promise. Credit growth was never the reason.

Source: DNB Financial Stability Report, spring 2022; DNB press releases of 27 May 2022 and 31 May 2023.
Huang & Gutiérrez · The Netherlands' CCyB

12/26

SAY

- What reasons did DNB actually give? In May 2022 it wrote that it was **restoring the buffers** it had lowered in March 2020. It acknowledged the uncertainty from the war, but said the robust recovery justified a gradual build-up.
- In May 2023 it went further. It wrote that the credit gap showed **no signs of excessive credit growth**. Instead it pointed to investors' rising risk appetite, falling property prices, the debt sustainability of firms and governments, and banks that were robust, partly thanks to higher rates.
- So both decisions rested on the same three pillars: a normal-to-elevated risk picture, strong banks and the 2020 promise. Credit growth was never the reason.

NUMBERS

- 1% step: about €3.3bn of extra CET1 (announced 25/27 May 2022)
- 2% step: about €3.4bn (announced 31 May 2023)

TERMS EXPLAINED

- **Cyclical systemic risk:** Risk that builds up over the financial cycle (credit, asset prices, risk-taking) and can materialise system-wide.

The buffer was meant for shocks nobody can forecast, while the known hot spots got their own tools

Q2 · CRITERIA & RISKS

Shocks nobody can forecast

war, energy, a sudden tightening of financial conditions, bank turmoil abroad (SVB, Credit Suisse)

CCyB

releasable whatever the source of the shock

The known hot spot: housing

an "overheated" market, high mortgage debt of first-time buyers, interest-only loans

Borrower limits, tax reform, mortgage risk-weight floor

(floor in force Jan 2022 – Nov 2026)

The size of the big banks

a banking sector of about 280% of GDP (end-2022)

O-SII buffer

structural, for systemically important banks

One instrument per target, as the Tinbergen principle suggests. If housing had been the target, the CCyB would be the wrong tool, because it raises the cost of all lending, not only mortgages.

Source: DNB Financial Stability Reports, spring 2022 and spring 2023; ESRB O-SII notification NL (31 May 2023); DNB (24 Apr 2026); Tinbergen (1952).
Huang & Gutiérrez · The Netherlands' CCyB

13/26

SAY

- Which risks was the buffer meant to cover? DNB matched each risk with its own tool.
- The top row is **shocks nobody can forecast**: war, an energy shock, a sudden tightening of financial conditions, or bank turmoil abroad like Silicon Valley Bank and Credit Suisse in 2023. These go to the countercyclical buffer, because it can be released whatever the source of the shock.
- **Housing**, the known hot spot, was handled by borrower limits, tax reform and a mortgage risk-weight floor. The **size of the big banks** is handled by the structural buffer for systemically important banks.
- This is the Tinbergen principle: one instrument per target. If housing had been the target, the countercyclical buffer would have been the wrong tool, because it raises the cost of all lending.

NUMBERS

- Banking sector: $\approx 400\%$ of GDP (when O-SII buffers were phased in) $\rightarrow 280\%$ (end-2022)
- Risk-weight floor in force Jan 2022 – Nov 2026

TERMS EXPLAINED

- **LTV / LTI**: Loan-to-value / loan-to-income. Caps on mortgage size relative to house value or income.
- **Tinbergen principle**: Reach each policy target with its own instrument.
- **Borrower-based measures**: Rules on borrowers (LTV, LTI) rather than on bank capital.

NEXT $\rightarrow$ "Now we can test the evidence against both readings."

Tested against the evidence, every prediction fits normalisation and none fits overheating

Q2 · CRITERIA & RISKS

	If DNB was normalising	If DNB was fighting a boom	What actually happened	
Credit indicators	the buffer ignores them	the buffer rises with them	the indicators fell while the buffer rose	✓
The path	+1pp a year, stop at 2%	follows the risk	1%, then 2%, then held	✓
2023: prices falling, rates jumping	raise anyway	pause	DNB raised	✓
2024–26: house prices up 10% again	stay at 2%	go above 2%	held at 2%	✓
Other buffers	cut structural buffers in return	keep them	cut in 2020 and 2024	✓

Five predictions, five matches for **normalisation**. The last row is the key to the next slide.

Source: BIS (WS_CREDIT_GAP, WS_DSR, WS_TC, WS_SPP); DNB press releases 2022–2026 (house prices: DNB, 26 Mar 2026); DNB framework (2022); ESRB O-SII notifications. Charts for each row: Appendix A1–A2.
Huang & Gutiérrez · The Netherlands' CCyB

14/26

SAY

- Now we can test the two readings. If DNB was fighting a boom, the buffer should follow credit and prices. If it was normalising, it should follow a fixed path back to its normal level. Row by row:
- **Credit indicators:** they all fell while the buffer rose. **The path:** one point a year and a stop at two, exactly as normalisation predicts. **2023:** prices falling and rates jumping; a boom fighter would pause, but DNB raised. **2024 to 2026:** house prices up about ten percent again; a boom fighter would go above two percent, but DNB held. **Other buffers:** normalisation means cutting structural buffers in return, and DNB did so twice.
- *(pause)* Five predictions, five matches for normalisation, and none for overheating. This is the core of our argument.

NUMBERS

- Gap, debt-service ratio and household debt ratio all falling in 2022–23
- House prices +10% y/y (DNB, Mar 2026) · CCyB held at 2% since May 2024

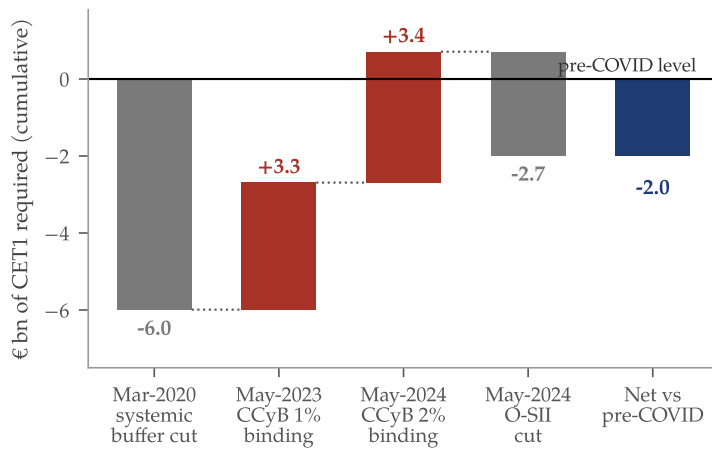
TERMS EXPLAINED

- **Phase 3 ("increased risk"):** The framework phase in which the CCyB goes above 2%.

NEXT → "And that last row tells us the refill was a swap, not a squeeze."

And the refill was a swap, not a squeeze: capital moved from structural buffers into the releasable one

Q2 · CRITERIA & RISKS



By our estimate, the 2020 cut released about €6bn and the 2024 cut another €2.7bn, while the two CCyB steps added €3.3bn and €3.4bn. Measured against the pre-COVID requirement, banks hold roughly the same amount of capital, or slightly less.

DNB described the swap as “more or less capital-neutral for the three large banks” (2020) and later as “a limited increase in the net capital requirements” (Knot, 2023).

Capital moved from buffers that cannot be released into one that can. Relative to 2021, however, it was a modest increase.

Source: Own calculation (cuts valued at end-2022 RWA of the four largest banks; CCyB amounts: DNB, sector-wide); DNB (17 Mar 2020); Knot (7 Jun 2023).
Huang & Gutiérrez · The Netherlands' CCyB

15/26

SAY

- And the last row shows that the refill was a swap, not a squeeze. By our estimate, the 2020 cut released about six billion euros, the two countercyclical steps added 3.3 and 3.4 billion, and the 2024 cut released another 2.7 billion. Net, compared with before COVID, banks hold roughly the same required capital, or slightly less. DNB called it more or less capital-neutral for the three large banks.
- Two caveats: the bases differ, so the numbers are approximate, and compared with 2021 the refill was a modest increase.
- So capital was **moved** from buffers that cannot be released into one that can. The total stayed similar; what changed is who can use it, and when.

NUMBERS

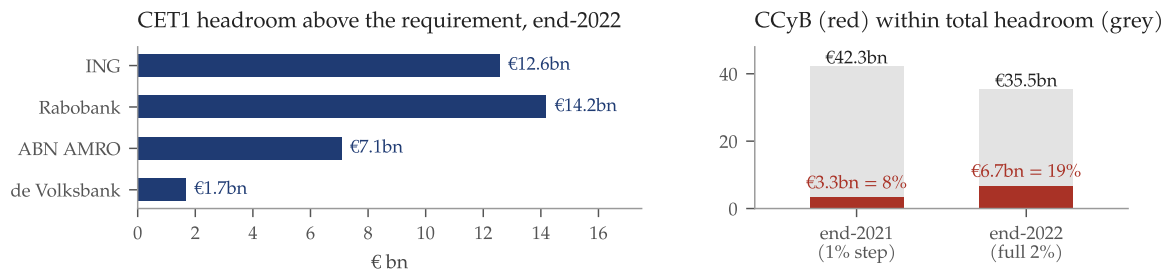
- -6.0 (2020: ING 1.7, Rabo 2.4, ABN 1.9) · $+3.3$ (May-23) · $+3.4$ (May-24) · -2.7 (May-24 O-SII) → net ≈ -2.0 €bn vs pre-COVID
- Valued at end-2022 total RWA; BNG excluded

TERMS EXPLAINED

- **Structural buffer:** A capital charge for permanent features (size, interconnectedness), kept through the cycle.
- **UK approach:** The UK offset its 2% neutral CCyB by lowering Pillar 2A and resolution requirements.

So was it the wrong moment? No. Higher rates gave banks the profits to build the buffer cheaply

Q3 · IMPLICATIONS



Headroom is the core capital a bank holds above its requirement. The four largest banks had €35.5bn of it at the end of 2022, so the full buffer of €6.7bn used at most a fifth. Higher net interest income meant the buffer could be built from retained earnings rather than by cutting loans, and the ECB finds that a gradual build-up by profitable banks keeps the costs low.

It was not the wrong moment. Arguably it was the right one.

Source: Bank disclosures (ING, Rabobank, ABN AMRO, de Volksbank, 2021–22), de Volksbank requirement derived; DNB FSR 2023; ECB Macropprudential Bulletin 24 (2024) and 31 (2025, Box 2).
Huang & Gutiérrez · The Netherlands' CCyB

16/26

SAY

- So, was it the wrong moment? We think not. The storm actually made it the **cheapest** moment.
- Headroom is the capital a bank holds above its requirement. At the end of 2022 Rabobank had 14.2 billion euros, ING 12.6, ABN AMRO 7.1 and de Volksbank 1.7, about 36 billion in total. The first step of 3.3 billion used eight percent of that, and the full buffer of 6.7 billion at most a fifth.
- And here is the twist. The same rising rates that made the moment look dangerous lifted bank profits, so banks could build the buffer from retained earnings instead of cutting loans. ECB research agrees that profitable banks and a gradual build-up keep the costs low.

NUMBERS

- Headroom end-2021 / end-2022 (€bn):
 - ING 16.9 / 12.6
 - Rabobank 15.7 / 14.2
 - ABN AMRO 7.9 / 7.1
 - de Volksbank 1.9 / 1.7
- Total headroom €42.3bn / €35.5bn
→ CCyB €3.3bn = 8%, €6.7bn = 19%

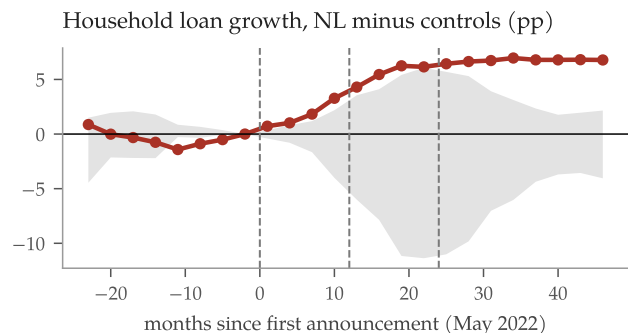
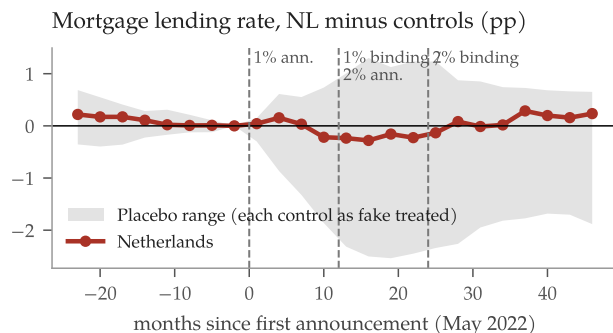
TERMS EXPLAINED

- **Headroom:** CET1 capital above the bank's requirement (its MDA trigger).
- **Retained earnings:** Profits kept in the bank instead of being paid out.

NEXT → "And borrowers did not pay either."

Borrowers did not pay either: credit kept flowing and its price moved with the rest of the euro area

Q3 · IMPLICATIONS



We compare the Netherlands with five euro-area countries that never changed their buffer (Austria, Finland, Italy, Malta, Luxembourg). The red line is the Dutch difference; the grey band shows differences that arise by chance when each control country is treated as if it had raised its buffer. Mortgage rates stayed inside the band, and household credit grew faster than in the controls, partly as a catch-up.

We find **no sign of contraction**, although this is not a causal estimate. DNB reached the same view in September 2026.

Source: ECB Data Portal (MIR, BSI); ESRB CCyB table (control selection); DNB press release 17 Sep 2026; own two-way fixed-effects event study with placebo inference.

Huang & Gutiérrez · The Netherlands' CCyB

17/26

SAY

- Did borrowers pay instead? We compare Dutch lending with five euro-area countries that never changed their buffer: Austria, Finland, Italy, Malta and Luxembourg.
- The red line is the Netherlands minus those countries. The grey band shows the differences that arise by chance, when we pretend each control country was the one that raised its buffer. Mortgage rates stay **inside the band** at all three dates. Household credit actually grew **faster**: Dutch bank credit grew 16.6 percent from early 2022 to early 2026, against 7.9 percent in the euro area, partly as a catch-up after lagging before.
- So we find **no sign of contraction**. With only one treated country this is not a causal estimate, but DNB reached the same conclusion in September 2026.

NUMBERS

- Dates: May-22 (1% announced) · May-23 (1% binding, 2% announced) · May-24 (2% binding)
- Bank credit 2022Q1–2026Q1: NL +16.6% vs euro area +7.9%; 2019–22: NL +4.3% vs +11.3%

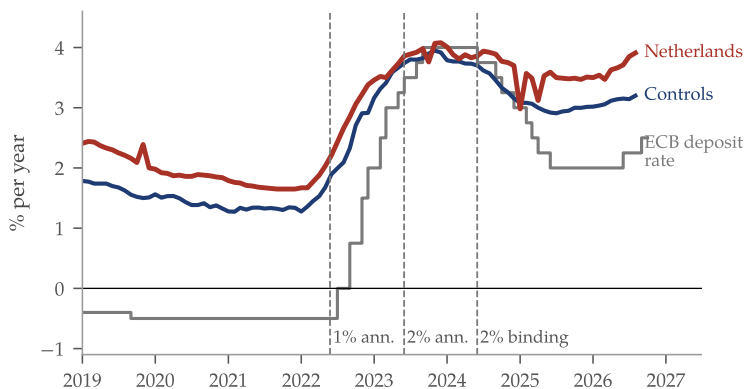
TERMS EXPLAINED

- **Event study:** Tracks the difference between treated and control groups before and after an event.
- **Placebo test:** Rerun the analysis pretending an untreated country was treated, to see what chance alone produces.

NEXT → HAND-OVER: "So banks and borrowers were fine. What about the wider effects? Zheng."

Nor did the buffer double the ECB's squeeze. It let monetary policy focus on inflation

Q3 · IMPLICATIONS



A natural worry is that two tightenings at once, higher policy rates and higher capital requirements, would choke credit. The chart shows that Dutch mortgage rates followed the ECB's rate and moved closely with those in the control countries.

The ECB argues that early CCyB activation "helps monetary policy focus on its primary objective of price stability." (ECB, 2025)

With one monetary policy for twenty countries, the CCyB is the Dutch-specific lever, and it is insurance in case a rate shock goes wrong.

Source: ECB Data Portal (MIR mortgage rates; deposit facility rate); ECB Macroprudential Bulletin 31 (Detken, Hempell & Pirovano, 2025). Controls: AT, FI, IT, MT, LU.

Huang & Gutiérrez · The Netherlands' CCyB

18/26

SAY

- Thank you, Diego. A natural worry is that two tightenings at once, higher policy rates and higher capital requirements, would choke credit. The chart shows that this did not happen. Dutch mortgage rates, in red, followed the ECB's deposit rate, in grey, and moved closely with mortgage rates in the control countries, in blue.
- The ECB argues that building buffers early lets monetary policy focus on its main goal, price stability. With one monetary policy for twenty countries, the countercyclical buffer is the **Dutch-specific lever**. It is also insurance in case a rate shock goes wrong.

NUMBERS

- ECB hiking from Jul 2022 (first +50bp) to at least Sep 2023

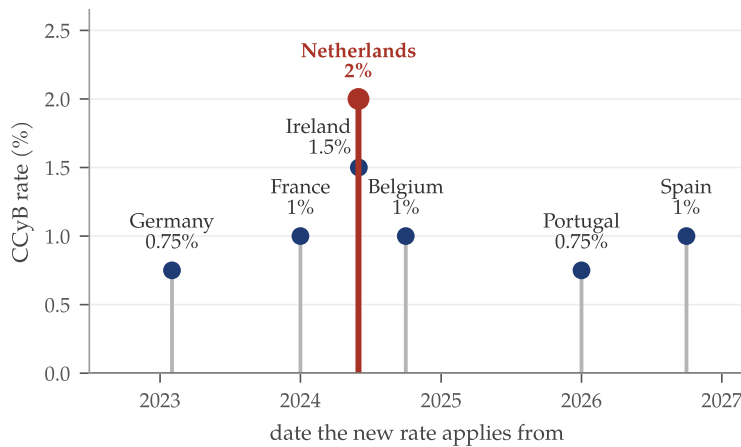
TERMS EXPLAINED

- **Macroprudential policy:** Policy aimed at the stability of the financial system as a whole.
- **Monetary policy:** Interest-rate policy aimed at price stability (ECB, euro area-wide).

NEXT → "Through reciprocity, the 2% also reaches foreign lenders."

Through reciprocity the 2% also binds foreign lenders, and the rest of Europe is moving the same way

Q3 · IMPLICATIONS



Under EU reciprocity, foreign banks that lend into the Netherlands must also hold the Dutch 2% on those loans. This limits leakage to foreign lenders, a known weakness of national capital rules, although lending can still move to non-banks.

The Netherlands was early and set the highest rate, but it was not alone. Since 2023 Germany, France, Ireland, Belgium, Portugal and Spain have all raised their buffers.

The Dutch move is part of a broader European shift towards keeping releasable capital in normal times.

Source: DNB press release 31 May 2023; ESRB CCyB rates table (Sep 2026); ECB Macroprudential Bulletin 21 (Behn et al., 2023); Aiyar, Calomiris & Wieladek (2014).
Huang & Gutiérrez · The Netherlands' CCyB

19/26

SAY

- The buffer also reaches across borders. Under EU reciprocity, foreign banks that lend into the Netherlands must also hold the Dutch two percent on those loans. This limits leakage to foreign lenders, which is a known weakness of national capital rules, although lending can still move to non-banks.
- And the Netherlands is part of a European shift. The chart shows the rates and the dates from which they apply. Germany, France, Ireland, Belgium, Portugal and Spain have all raised their buffers since 2023. The Netherlands was early and set the highest rate, but it was not alone.

NUMBERS

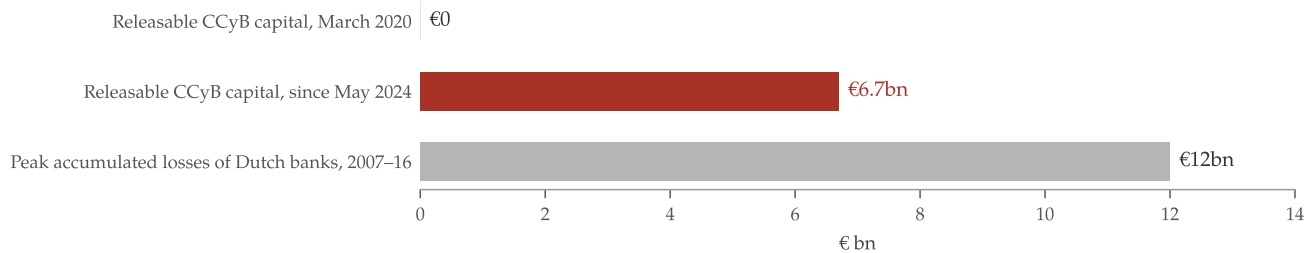
- Germany 0.75% (Feb 2023) · France 1.0% (Jan 2024) · Ireland 1.5% (Jun 2024) · NL 2.0% (May 2024) · Belgium 1.0% (Oct 2024) · Portugal 0.75% (Jan 2026) · Spain 1.0% (Oct 2026)
- Positive-neutral rates in 2023: Cyprus 0.5%, Estonia and Lithuania 1.0%, Ireland 1.5%, NL 2.0%

TERMS EXPLAINED

- **Reciprocity:** Foreign banks apply the host country's CCyB rate to exposures there (mandatory in the EU up to 2.5%).
- **Leakage:** Credit shifting to lenders not covered by the rule (foreign branches, non-banks).

The result: next time, DNB can release €6.7bn overnight

Q3 · IMPLICATIONS



In March 2020 the Netherlands had no releasable capital. Today it has €6.7bn, about half of the peak losses after the financial crisis and enough to support up to €150bn of lending. Research suggests the payoff is asymmetric: a one-point increase in normal times reduces lending by only about 0.1%, while a one-point release in bad times can raise it by up to 10%.

With the mortgage risk-weight floor expiring in November 2026, DNB says this “underpins the importance of . . . the current countercyclical capital buffer (CCyB) of 2%.”

The buffer is cheap to hold and powerful to release.

Source: DNB framework (2022); Lang & Menno (2023), ECB WP 2828; DNB press release 24 Apr 2026.
Huang & Gutiérrez · The Netherlands' CCyB

20/26

SAY

- The result is shown in this chart. In March 2020 the Netherlands had no releasable capital at all. Today it has **6.7 billion euros**, about half of the peak losses after the financial crisis, and enough to support up to 150 billion euros of lending.
- This buffer is cheap to hold and powerful to release. Research on European banks suggests that a one-point increase in normal times reduces lending by only about 0.1 percent, while a one-point release in bad times can raise lending by up to ten percent.
- And its role is growing. The floor on mortgage risk weights expires in November 2026, and DNB says that this underlines the importance of the two-percent buffer.

NUMBERS

- €0 → €6.7bn releasable · ≈ half of €12bn peak losses · up to €150bn of lending
- Lang & Menno (2023): –0.1% lending per +1pp (normal times) vs up to +10% per 1pp release (bad times)

TERMS EXPLAINED

- **State-dependent effect:** The impact of a policy depends on the state of the economy (small in good times, large in bad times).

The price is reliance on discretion, and the buffer has never been tested

Q3 · IMPLICATIONS

Credibility	Without a mechanical rule, markets must trust DNB's judgement, so DNB has to explain every decision. This is the classic trade-off between rules and discretion (Kydland & Prescott, 1977).
Calibration	Why 2% and not 1.5%? The ECB's loss-based estimates range from 1.1% to 1.8%, so the Dutch rate sits at the top of the banking union.
Untested	The buffer has never been released. Will banks lend the freed capital, or hoard it, as many did with usable buffers in COVID?
Baseline	Compared with 2021 rather than 2019, the refill <i>was</i> an increase: €3.3bn before the O-SII offset.
The same profits, twice	The ECB lists the Netherlands among the countries that adopted levies on banks' excess profits in 2022–23. These levies partly claim the earnings that fund the buffer.

None of these overturns the verdict, but they define what DNB must get right next time.

Source: Kydland & Prescott (1977); De Nora et al. (2025); ECB Macroeprudential Bulletin 21 (2023) and 31 (2025); DNB (2022, 2023).
Huang & Gutiérrez · The Netherlands' CCyB

21/26

SAY

- Of course there is a price, and we see five critiques.
- **Credibility:** without a rule, markets must trust DNB's judgement, so DNB has to explain every decision. **Calibration:** why two percent, when the ECB's estimates range from 1.1 to 1.8? **Untested:** the buffer has never been released, and in 2020 many banks did not use the buffers they were allowed to use. **Baseline:** compared with 2021 rather than 2019, it was an increase. **Profits:** the same profits that fund the buffer were also targeted by levies on excess profits.
- None of these overturns our verdict, but they define what DNB must get right next time.

NUMBERS

- ECB loss-based range 1.1–1.8% vs NL 2% · +€3.3bn vs 2021 before the O-SII offset

TERMS EXPLAINED

- **Rules vs discretion:** The trade-off between predictable formulas and flexible judgement (Kydland & Prescott 1977).
- **Excess-profit levy:** A temporary tax on bank profits seen as windfalls from higher rates.

In short, the promise was kept: the buffer was refilled in a storm, at the moment it was cheapest

CONCLUSION

Q1 · ENVIRONMENT

A recovered economy was hit by war, inflation and rate hikes. Credit was cold, housing was hot and then cooled, and the banks were strong and profitable.

Q2 · CRITERIA AND RISKS

A 2% normal level, calibrated on past losses, insures against shocks nobody can forecast. Housing went to other tools, and structural buffers were cut in return.

Q3 · IMPLICATIONS

The refill was cheap and largely a swap, with no sign of contraction. It complements monetary policy, binds foreign lenders and leaves €6.7bn releasable, at the price of discretion.

In March 2020, the Netherlands had nothing to release.
Next time, it will.

The best time to fill a buffer is when nobody thinks you need it, even in a storm.

Huang & Gutiérrez · The Netherlands' CCyB

22/26

SAY

- Let us conclude with the three questions of the assignment. The **environment** was a recovered economy hit by war, inflation and rate hikes, with cold credit, a cooling housing market and strong banks. The **criteria** were a two-percent normal level calibrated on past losses, as insurance against shocks nobody can forecast. The **implications**: the refill was cheap, largely a swap, with no sign of contraction, and it leaves 6.7 billion euros ready to release, at the price of more discretion.
- *(pause)* In March 2020 the Netherlands had nothing to release. **Next time, it will.** The best time to fill a buffer is when nobody thinks you need it, even in a storm. Thank you, and we look forward to your questions.

NUMBERS

- 0% → 2% · −33 / −47pp gap · ≤ 1/5 of headroom · €6.7bn releasable

TERMS EXPLAINED

- **Normalisation**: Returning the buffer to its normal-times level, not reacting to overheating.

Glossary

APPENDIX

CCyB	Countercyclical capital buffer: 0–2.5% of risk-weighted assets on domestic exposures; raised with a 12-month lag, released immediately.	Credit-to-GDP gap	Credit/GDP ratio minus its long-run trend; the Basel reference for the CCyB (gap 2–10pp → buffer 0–2.5%).
Positive neutral CCyB	A CCyB kept above zero when risks are neither high nor low, so it can be released for any shock.	HP filter	Hodrick-Prescott filter: estimates a smooth trend; one-sided uses only past data (real time), two-sided also future data.
CET1	Common Equity Tier 1: the highest-quality capital (shares, retained earnings), as % of RWA.	Headroom	CET1 capital above a bank's requirement / MDA trigger.
RWA	Risk-weighted assets: assets weighted by their riskiness (e.g. mortgages count less than corporate loans).	Reciprocity	Foreign banks must apply the Dutch CCyB rate to their Dutch exposures.
Pillar 1 / P2R / P2G	Legal minimum (4.5% CET1) / bank-specific add-on set by the supervisor / non-binding supervisory guidance.	Borrower-based measures	Limits on loan-to-value (LTV) and loan-to-income (LTI) of new mortgages.
CCoB	Capital conservation buffer, 2.5% for all banks.	Risk-weight floor	Minimum average risk weight on Dutch mortgages of banks using internal models (Art. 458 CRR), Jan 2022 – Nov 2026.
O-SII / SRB	Buffer for other systemically important institutions / systemic risk buffer (structural, not designed for release).	Event study / placebo	Compare NL with countries that did not change their CCyB around each date / repeat with each control as a fake "treated" country.
MDA	Maximum distributable amount: if CET1 falls below the combined buffer requirement, payouts are automatically restricted.	pp / bp	Percentage point / basis point (0.01pp).

Huang & Gutiérrez · The Netherlands' CCyB

23/26

SAY

- Backup only. Use it when a question involves a term; point to the row and give the one-line definition.

NUMBERS

- (definitions only)

TERMS EXPLAINED

- All key terms of the talk are on the slide

Methods and references

APPENDIX

Methods. Credit gap: one-sided (recursive) and two-sided HP filter, $\lambda = 400,000$, sample from 1961Q1; Hamilton (2018) regression filter, $h = 8, 20, p = 4$. Headroom: (CET1 ratio – CET1 requirement/MDA) $\times$ total RWA per bank; CCyB amount sector-wide (upper bound); 6–47% assumption grid as sensitivity. Event study: two-way FE, NL $\times$ 3-month event-time bins, country and month FE; placebo inference over controls with no CCyB change 2021–25. Code and data: GitHub repository (code/, data/).

Primary sources

BCBS (2024). Range of practices in implementing a positive neutral CCyB (d585).
 DNB (2020). DNB lowers bank buffer requirements to support lending, 17 Mar.
 DNB (2022). Analytical framework for setting the CCyB in the Netherlands.
 DNB (2022, 2023). CCyB press releases, 27 May 2022, 31 May 2023; O-SII news item, 31 May 2023.
 DNB (2026). CCyB decisions 26 Mar and 17 Sep; risk-weight measure expires, 24 Apr.
 DNB. Financial Stability Report, spring 2022 and spring 2023.
 Knot, K. (2023). Introductory statement, Standing Parliamentary Committee for Finance, 7 Jun.
 ECB (2020). Temporary capital and operational relief, 12 Mar; ECB (2022). Monetary policy decisions, 21 Jul.
 ESRB. CCyB rates table; O-SII notifications NL (27 Nov 2020; 31 May 2023); COVID-19 measures.
 Data: BIS (credit gaps, total credit, DSR, property prices); ECB Data Portal (MIR, BSI); CBS; bank annual and Pillar 3 reports (ING, Rabobank, ABN AMRO, de Volksbank).

Literature

Aiyar, Calomiris & Wieladek (2014). Does macro-prudential regulation leak? *JMCB* 46(s1).
 Behn, Pereira, Pirovano & Testa (2023). A positive neutral rate for the CCyB. *ECB MPB* 21.
 Couaillier, Lo Duca, Reghezza & Rodriguez d'Acari (2022). Caution: do not cross! *ECB WP* 2644.
 De Nora, Pereira, Pirovano & Stammwitz (2025). From losses to buffer. *ECB WP* 3061.
 Detken, Hempell & Pirovano (2025). Macroprudential and monetary policy interaction. *ECB MPB* 31.
 Drehmann, Borio, Gambacorta, Jiménez & Trucharte (2010). Countercyclical capital buffers: exploring options. *BIS WP* 317.
 Edge & Meisenzahl (2011). The unreliability of credit-to-GDP ratio gaps in real time. *IJCB* 7(4).
 Hamilton (2018). Why you should never use the Hodrick-Prescott filter. *REStat* 100(5).
 Herrera, Scalone & Pirovano (2024). The importance of being positive. *ECB MPB* 24.
 Kydland & Prescott (1977). Rules rather than discretion. *JPE* 85(3).
 Lang & Menno (2023). The state-dependent impact of changes in bank capital requirements. *ECB WP* 2828.
 Repullo & Saurina (2011). The countercyclical capital buffer of Basel III: a critical assessment. *CEMFI WP* 1102.
 Tinbergen (1952). *On the Theory of Economic Policy*. North-Holland.

Huang & Gutiérrez · The Netherlands' CCyB

24/26

SAY

- Backup for method questions.
 - **Credit gap:** one-sided and two-sided HP filter ($\lambda = 400,000$), plus Hamilton regression filter.
 - **Headroom:** (CET1 ratio minus requirement) $\times$ total RWA, per bank.
 - **Event study:** two-way fixed effects, three-month bins, placebo inference over five never-treated countries.
- All references are verified.

NUMBERS

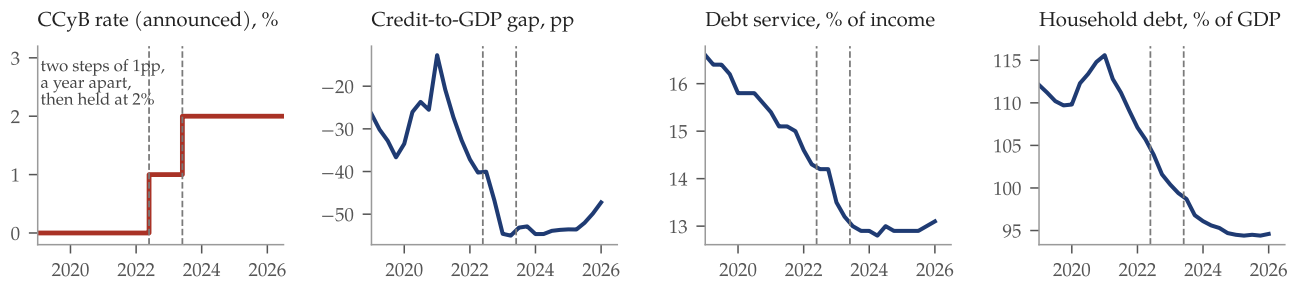
- Data: BIS, ECB Data Portal (MIR, BSI), ESRB, DNB, bank reports

TERMS EXPLAINED

- **Two-way fixed effects:** Regression with country and time dummies, so differences are measured relative to common trends.

Appendix A1 · The verdict, rows 1 and 2: the buffer rose in two fixed steps while every credit indicator fell

APPENDIX



Row 2 · THE PATH

Normalisation predicts one percentage point a year and a stop at 2%. DNB announced 1% in May 2022 and 2% in May 2023, and has held 2% ever since.

The buffer moved on a timetable, not with credit.

Row 1 · CREDIT INDICATORS

A boom fighter raises the buffer when credit indicators rise. Between the two announcements (dashed lines) the gap, the debt-service ratio and household debt all fell.

Source: ESRB CCyB rates table; DNB press releases (27 May 2022, 31 May 2023); BIS (WS_CREDIT_GAP, WS_DSR, WS_TC). Dashed lines: CCyB announcements.
Huang & Gutiérrez · The Netherlands' CCyB

25/26

SAY

- (Only if asked about page 14.) The left panel shows the path: two steps of one point, a year apart, then held at two percent, exactly as normalisation predicts.
- The other three panels show the credit indicators. Between the two announcements the credit gap, the debt-service ratio and household debt all fell, so the buffer did not follow credit.

NUMBERS

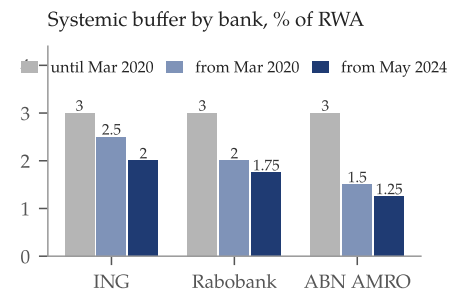
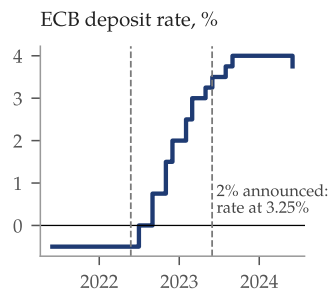
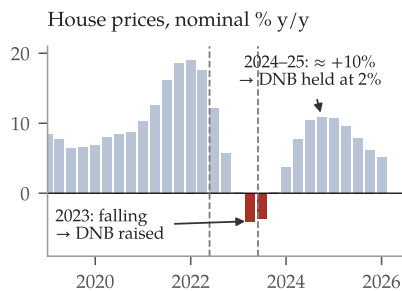
- CCyB announced: 1% (25 May 2022), 2% (31 May 2023); held at 2% since
- Gap -33pp (2021Q4) $\rightarrow -47\text{pp}$ (2022Q4); household debt 107% (2022Q1) $\rightarrow$ 100% (2023Q1) of GDP; debt-service ratio 14.6% (2022Q1)

TERMS EXPLAINED

- **Debt-service ratio:** Interest plus principal payments as a share of income.

Appendix A2 · The verdict, rows 3 to 5: DNB raised into falling prices, held when prices boomed, and cut other buffers

APPENDIX



Row 3 · 2023

Prices were falling and the ECB rate stood at 3.25%. A boom fighter would pause; DNB raised.

Row 4 · 2024–26

House price growth returned to about 10% a year. A boom fighter would go above 2%; DNB held at 2%.

Row 5 · OTHER BUFFERS

DNB cut the large banks' systemic buffers in 2020 and again in 2024.

All five rows point the same way: **normalisation, not tightening.**

Source: BIS (WS_SPP); ECB Data Portal (deposit facility rate); ESRB O-SII notifications (2020, 2023); DNB (17 Mar 2020).
Huang & Gutiérrez · The Netherlands' CCyB

26/26

SAY

- (Only if asked about page 14.) On the left, house prices: in 2023 they were falling, yet DNB announced two percent. In 2024 and 2025 they grew about ten percent a year, yet DNB held at two percent.
- In the middle, the ECB deposit rate stood at 3.25 percent when the second step was announced.
- On the right, the systemic buffers of the three large banks were cut in 2020 and again in 2024, which is what normalisation predicts.

NUMBERS

- House prices, nominal y/y (BIS): –4.0% (2023Q2); +10.8% (2024Q4 and 2025Q1); +5.1% (2026Q1)
- ECB deposit rate 3.25% on 31 May 2023
- Systemic buffers: ING 3 → 2.5 → 2.0%; Rabobank 3 → 2.0 → 1.75%; ABN AMRO 3 → 1.5 → 1.25% (until Mar 2020 / from Mar 2020 / from May 2024)

TERMS EXPLAINED

- **O-SII buffer:** Structural capital charge on systemically important banks; replaced the systemic risk buffer from 29 Dec 2020.